

You don't have to choose your career today.

But how do you choose
a career **you've never seen?**

Career as an AI Engineer

What AI engineers actually do, how you can become one, and what to learn first.

AUDIENCE

Classes IX to XII

VENUE

KV RK Puram, Sector 2

DEPARTMENT

DEPFE, NCERT, New Delhi

Tushar Gaurav

Software Engineer

Orangewood Labs, Noida, India

tushgaurav.com

B.Tech in Computer Science Engineering

Noida Institute of Engineering and Technology, 2020 to 2024

I build the software that runs robots, and AI is a big part of it.

Full Stack

Cloud & DevOps

AI



Scan to visit my website

Before you start with AI

AI doesn't work on its own. It lives inside apps and websites you already use every day. So let's first understand, in simple words, how those are built.

01

FOUNDATIONS

How does an app **actually** work?

Think of what happens when you open Instagram or order food on Zomato. Six words explain almost all of it.

Client

Your phone or laptop. The device that asks for something, like “show me my feed”.

Server

A powerful computer far away that hears your request and sends back an answer.

Frontend

Everything you can see and tap: screens, buttons, menus and forms.

Backend

The brain behind the scenes. It checks your password, finds your order, saves your photo.

Database

A giant, well-organised store of information: usernames, posts, marks, notices.

Cloud

Renting computers over the internet instead of buying your own, so an app can run 24×7 for millions of people.

Every AI engineer is a **software engineer** first

An AI model on its own is like a car engine without a car. Someone has to build the rest: the screens, the buttons, the data. That is software development.

Learn to code

Pick one language and get comfortable. **Python** is the friendliest start, and it is also the language of AI. Build tiny things: a calculator, a quiz, a to-do list.

Know how apps are built

Understand how the frontend, backend and database talk to each other. Build one small website end to end, even an ugly one.

Get used to the tools

Git to save your work, the terminal, and putting a project on the internet (the cloud). Most of all, learn to search for an error and fix it yourself.

People who build whole apps like this are called **Full Stack Developers**: a great career on its own, and the most common doorway into AI.

Artificial Intelligence Engineer

Someone who builds software that learns from data and makes decisions: recommending your next video, understanding your voice, or spotting a disease in a medical scan.

- 01** **Myths and how to get in**
What people get wrong, and the study routes
- 02** **What the work is like**
Day-to-day tasks and tools
- 03** **Working life and personality**
Where you work, and who enjoys it
- 04** **Growing in the career**
From first job to leadership, and what it pays
- 05** **Where to work and study**
Companies, research labs and colleges
- 06** **What to do from today**
Courses, projects and the future

Myths about AI

Myth: AI is always right

AI can sound very confident and still be wrong. Someone has to test it, check the facts and fix it. That someone is the AI engineer.

Myth: AI will take every job

AI takes over tasks, not whole jobs. Judgement, responsibility and knowing a field deeply still matter, and people who understand AI have the edge.

Myth: AI engineering is just typing prompts

Prompting ChatGPT is one small part. Building AI that works reliably needs coding, handling data, testing, and running it for real users.

Myth: You need an "AI degree" to work in AI

Many roads lead here: Computer Science, Maths, Statistics, IT, or any engineering degree plus self-study. What you build matters more than the label.

Routes into **AI engineering**

Most people start with 10+2 in Science with Physics, Chemistry and Maths. Of these, Maths matters the most.

Degree or diploma

- A bachelor's (**B.Tech / B.E. / B.Sc / BCA**) in **AI, Computer Science, IT, Maths or Statistics**
- Or a **diploma in AI and Machine Learning**
- Or any of these degrees plus **online AI courses**. Many working AI engineers learned this way

Study further (optional)

- A master's after your bachelor's: **M.Tech / M.Sc / MCA** in AI, Machine Learning, Data Science, Robotics or CS
- Helpful if you want to do research or work on the hardest problems
- Not compulsory. Plenty of people join companies straight after a bachelor's

Admission usually goes through entrance exams such as JEE Main / Advanced, or state exams like WBJEE. Always check the latest eligibility, fees and duration on the college's own website.

A typical week **and the tools**

A rough picture. It changes from team to team and product to product.

What you actually do

- Collect and clean data. Honestly, a big part of the job is here
- **Train models:** teach the computer to learn patterns from that data
- Put the model inside a real app, then **keep watching that it works**
- Work on chatbots, recommendations, computer vision and **large language models** (the tech behind ChatGPT)
- Write clean code, run experiments, and read about new ideas

Tools you will use

- **Languages.** Python first, plus SQL for working with data
- **AI libraries.** PyTorch, TensorFlow, scikit-learn, Hugging Face
- **Cloud and tools.** AWS, Google Cloud or Azure, Docker, Git
- **Data.** Pandas, NumPy, vector databases
- **People.** You work with data scientists, designers and other software engineers

Work environment **and** personal traits

Hours and setup

- Usually an **office or hybrid** setup
- Around **8 hours a day, 5 days a week**
- Some late nights before a big launch or deadline

Who you work with

- **Data scientists, software engineers** and product teams
- Lots of discussion, whiteboards and reviewing each other's code
- Inclusive field: opportunities exist for people with disabilities

You'll enjoy it if you...

- Like maths and science
- Enjoy puzzles and figuring out why something broke
- Are curious about how people think and behave
- Are patient. Experiments often fail before they work

Scholarships and education loans

Money should not stop you. There is a lot of help available if you look for it early.

Scholarships

- Most colleges give **merit scholarships**. Ask the admissions office directly
- Programmes to look up: IndusInd Foundation, Baba Gurbachan Singh, Samsung Star Scholar, NTPC and Foundation for Excellence
- **National Scholarship Portal**: one place for central, UGC / AICTE and state schemes

Education loans

- **Vidya Lakshmi**: compare, apply for and track education loans from many banks in one place
- Banks give education loans; the terms depend on the bank and the course
- Some states, including West Bengal, Odisha and Bihar, have student-credit schemes

Schemes and deadlines change every year. Check the current details before applying.

From your first job to leadership

Junior Engineer

Your first job. You learn from seniors and build features with guidance.

Senior Engineer

You own bigger pieces of the system and start helping the juniors.

Principal Engineer / Engineering Manager

You choose a path: go deep in technology, or lead a team of people.

Vice President

Run large engineering groups and decide what gets built.

Chief Technology Officer

Set the technology direction for the whole company.

A rough map, not a fixed timetable. Titles and steps vary from company to company.

Course fees and expected income

Approximate course fee

₹43,400 – ₹10,00,000

- Roughly the **total fee** for a course, not per year
- Varies a lot between government and private colleges
- Always check the full fee structure before enrolling

Approximate monthly income

₹42,000 – ₹1,33,333

- **Per month.** Depends on your experience, the company and the city
- Freshers start near the lower end; top companies pay well above this range
- Not a promise, just a picture

These are indicative figures, not a live salary survey. Compare with current job listings when the time comes.

Who hires **AI engineers**?

Almost everyone now. Here are the three big groups.

Tech companies and startups

From Google and Microsoft to small startups building apps, tools and robots (that is where I work).

Research labs

Government labs such as ISRO and DRDO, university research groups, and private research teams working on brand-new ideas.

Every other industry

Hospitals, banks, e-commerce, farming, education. Every field is adding AI, and they all need people who can build it.

Colleges to explore

A few well-known names

- **Indian Institute of Science**, Bengaluru, Karnataka
- **University College of Engineering, Osmania University**, Hyderabad, Telangana
- **Indian Institute of Technology Hyderabad**, Telangana
- Also worth a look: **IITs, NITs, IIITs** and good state universities with strong Computer Science departments

Before you pick a college

- This is a **starting point**, not a ranking
- Look at the actual course and the subjects it teaches
- Check entrance exams, fees and duration on the college's own website
- A good teacher and a strong coding culture matter more than a famous name

You can start **right now**

Free things to learn today

- Learn **Python**: YouTube, freeCodeCamp and Kaggle Learn are free
- DeepLearning.AI courses by Andrew Ng, once you know some Python
- **Kaggle**: practise on real data and join beginner competitions
- Google, AWS and Azure beginner certificates, later in college

Projects and internships

- Build small projects and put them on **GitHub**
- In college, try Google Summer of Code and company internships
- Research internships at IITs and IISc
- Share what you build: a blog post, a video, a demo

What the future looks like

- **Generative AI** and applied machine learning will keep growing
- Strong basics in coding and maths never go out of date
- Learn about **responsible and ethical AI**
- Tools change every year. The habit of learning is the real skill

Questions & discussion

There are no silly questions. Ask me anything about AI, coding, college or my job.

Thank You